# Testing Implementations of Human Debiasing Prompts in LLMs: Evidence from 22,773 Anchoring Bias Trials

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*See Authorship Statement for rationale.*

February 2026

**Abstract**

Can prompts that implement human debiasing techniques reduce anchoring bias in LLMs? We tested four techniques from cognitive psychology—Devil's Advocate, SACD (Self-Administered Cognitive Debiasing), Premortem, and Outside View—across 22,773 trials on 10 models.

**Finding:** Transfer is unreliable. Some prompts help: our SACD prompt achieves 93.7% of unanchored baseline (cross-model aggregate; see Methods for design caveats). But others backfire: our Devil's Advocate prompt produces responses *further from baseline than no intervention at all.* Standard evaluation metrics (susceptibility) would recommend the worst-performing prompt.

**Variance is severe.** SACD ranges from 48% to 128% of baseline depending on model. Rankings shift across domains—in 3 of 6 domains, no debiasing intervention outperforms the baseline. A prompt that works on one model may fail on another.

**Metrics matter.** We introduce MADB (Mean Absolute Deviation from Baseline) to complement aggregate metrics. SACD's 93.7% aggregate masks 18.1% per-trial error because overshoots and undershoots cancel. Aggregate metrics can hide poor performance.

**Recommendation:** Don't assume a prompt implementation transfers across models or tasks. Test per-model. Report MADB alongside aggregates. The standard metric (susceptibility) measures consistency, not accuracy—it can recommend the wrong prompt.

## 1 Introduction

Human debiasing techniques—Devil's Advocate, Premortem, deliberative reasoning—are increasingly applied to LLMs. But do specific prompts implementing these techniques actually work? We test one prompt per technique across 22,773 trials and find transfer is not straightforward: our SACD prompt's effectiveness varies dramatically across models (Table 5). Different prompt formulations may yield different results. Most concerning, standard evaluation metrics can recommend the wrong prompt.

### 1.1 Two Metrics, Two Rankings

Prior work evaluates debiasing using **susceptibility**—the gap between high-anchor and low-anchor responses:

$$\text{Susceptibility} = |\bar{R}_{high} - \bar{R}_{low}| \tag{1}$$

Lower gap = less susceptible. But susceptibility measures *consistency*, not *accuracy*. A prompt can achieve low susceptibility while landing far from the unanchored judgment.

**Our approach:** We add **% of baseline**—proximity to the model's unanchored judgment:

$$\% \text{ of Baseline} = \frac{R_{prompt}}{R_{baseline}} \times 100\% \tag{2}$$

where $R_{prompt}$ is the mean response under the technique (averaged across anchor conditions and trials) and $R_{baseline}$ is the model's mean unanchored response. 100% = the model's anchor-free response (which may itself be incorrect).

### 1.2 Contributions

(1) **Aggregate metrics mask bidirectional error.** *(Note: SACD uses ∼6 turns vs. 3 for others, confounding comparisons; see Section 6.4.)* Standard % of baseline can hide per-trial variance when overshoots and undershoots cancel (Section 5). We recommend reporting MADB alongside aggregates to reveal this hidden variance. (2) **Main study:** Deep validation (14,152 judicial trials, 10 models) showing substantial model-specific variance. (3) **Multi-domain extension:** Cross-domain evaluation (8,621 trials, 6 domains, 5 models) showing prompt rankings vary by domain and metric. Combined total: 22,773 trials.

## 2 Related Work

### 2.1 Anchoring Bias in Human Judgment

Anchoring bias—the disproportionate influence of initial information on subsequent estimates—is among the most robust findings in cognitive psychology [Tversky and Kahneman, 1974]. Even experts are susceptible: Englich et al. [2006] demonstrated that experienced judges' sentencing decisions were influenced by random numbers generated by dice rolls. Effect sizes of $d = 0.6$–$1.2$ persist regardless of anchor source or participant awareness. Our experimental paradigm adapts this judicial sentencing design.

### 2.2 Cognitive Biases in LLMs

LLMs exhibit human-like cognitive biases [Binz and Schulz, 2023, Jones and Steinhardt, 2022, Chen et al., 2025], including anchoring [Huang et al., 2025, Song et al., 2026]. Unlike humans, LLMs enable exhaustive systematic testing.

### 2.3 Debiasing Techniques

We test one prompt per approach for four debiasing techniques plus a control: **SACD** (iterative self-correction [Lyu et al., 2025]), **Devil's Advocate** (argue against initial response), **Premortem** (imagine decision failed [Klein, 2007]), **Random Control** (neutral filler to isolate turn-count effects), and **Outside View** (reference class forecasting [Kahneman and Lovallo, 1993, Sibony, 2019], excluded from rankings due to confound—see Section 6.3). Recent work on framing debiasing [Lim et al., 2026] shares conceptual overlap.

**Prompt design:** Prompts were derived from canonical descriptions in the cognitive bias literature (SACD from Lyu et al. [2025]; Premortem from Klein [2007]; Outside View from Kahneman and Lovallo [1993]). Devil's Advocate and Random Control were designed ad hoc to operationalize

the technique concept. No prompt optimization or iteration was performed—results reflect single implementations. Exact wording in Appendix A.

### 2.4 Evaluation Methodology

Standard anchoring evaluation uses **susceptibility** (high–low spread). We add **% of baseline** (proximity to unanchored judgment). Unlike the classic Anchoring Index [Jacowitz and Kahneman, 1995], which uses baseline in both numerator and denominator, our metric avoids circularity. Low Anchoring Index with poor baseline proximity indicates consistent but wrong responses—the failure mode we observe with our DA implementation. We use free-form numeric responses rather than MCQs to avoid position and label biases documented in benchmark evaluations [Nowak et al., 2026].

## 3 Methodology

### 3.1 Evaluation Metrics

We compare susceptibility (standard) with % of baseline (Section 1.1). Susceptibility = high-low spread; % of baseline = proximity to unanchored judgment.

**Mean Absolute Deviation from Baseline (MADB):** We define MADB as the average per-trial deviation from the model's unanchored baseline:

$$\text{MADB} = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{r_i}{b_m} - 1\right| \times 100\% \tag{3}$$

where $r_i$ is the response for trial $i$, $b_m$ is model $m$'s unanchored baseline response, and the sum is over all anchored trials. Lower MADB indicates responses closer to unanchored judgment. Unlike aggregate % of baseline, MADB does not allow positive and negative deviations to cancel. *Note:* Because deviations are computed relative to each model's own baseline, MADB aggregates meaningfully across models with different absolute baselines.

**Why both?** Metrics can give divergent rankings—reducing spread may move responses away from baseline (Section 4.2).

**Baseline as reference, not ground truth.** Following Jacowitz and Kahneman [1995], we measure deviation from each model's unanchored baseline rather than external ground truth. This operationalizes debiasing as *consistency*: similar outputs regardless of anchors. Baselines are not normatively correct (models disagree substantially, Table 2)—we measure anchor-independence, not correctness.

**Proportional anchors.** Anchors set at ±50% of each model's baseline ensure comparable pressure across models. Not circular: baselines measured independently, then used for anchors. 100% = anchor-independent judgment.

### 3.2 Experimental Design

#### 3.2.1 Models

We evaluated 10 models across 5 providers:

| Provider | Models |
|---|---|
| Anthropic | Claude Haiku 4.5, Sonnet 4.6, Opus 4.6 |
| OpenAI | GPT-4.1, GPT-5.2, o3, o4-mini |
| DeepSeek | DeepSeek-v3.2 |
| Others | Kimi-k2.5 (Moonshot), GLM-5 (Zhipu) |

#### 3.2.2 Conditions

1. **Baseline**: Sentencing prompt with no anchor
2. **Low anchor**: Prosecutor demand at baseline $\times$ 0.5
3. **High anchor**: Prosecutor demand at baseline $\times$ 1.5
4. **Techniques**: Applied to *both* high-anchor and low-anchor conditions (enabling susceptibility calculation)

#### 3.2.3 Techniques Evaluated

| Technique | Description |
|---|---|
| Outside View | "What typically happens in similar cases?" (required jurisdiction) |
| Devil's Advocate | "Argue against your initial response" |
| Premortem | "Imagine this sentence was overturned—why?" |
| Random Control | Extra conversation turns with neutral content |
| SACD | Iterative self-administered cognitive debiasing |

#### 3.2.4 Temperature Conditions

Tested at t=0, 0.7, 1.0. Descriptive analysis suggests no meaningful temperature$\times$technique interaction ($F(6, 8944) = 1.42$, $p = 0.203$; denominator df likely inflated by model clustering); temperature effects <3pp (percentage points), $d < 0.15$ (negligible relative to technique effects of $d = 0.5$–$1.0$). Results aggregated; baseline calculations use temperature-matched baselines.

#### 3.2.5 Trial Counts and Procedure

- **Total**: 14,152 judicial + 8,621 multi-domain = 22,773 analyzed
- **Per cell**: 30–90 trials (min $n = 30$ pre-specified; added trials for high-variance conditions)
- **Baselines**: 909 trials ( 90 per model)
- **Extraction**: Regex for integer months; 99.9% success
- **Anchors**: Proportional (high = 1.5$\times$baseline, low = 0.5$\times$)

Table 1: Trial distribution. Total unique trials: 14,152. Outside View is included in this count but excluded from prompt rankings due to confound (Section 6.3). Sample sizes shown are for primary analyses; prompt comparisons use matched model-temperature subsets.

| Condition | $n$ (analysis) |
|---|---|
| *Debiasing Techniques* | |
| SACD | 2,397 |
| Outside View | 2,423 |
| Random Control | 2,215 |
| Premortem | 2,186 |
| Devil's Advocate | 2,166 |
| *Control Conditions* | |
| Anchored (no debiasing) | 1,864 |
| Baseline (no anchor) | 909 |

#### 3.2.6 Statistical Analysis

Percentile bootstrap (10,000 resamples, trial-level sampling stratified by model—each resample maintains the original proportion of trials from each model to preserve the nested structure) for CIs. **CI caveat:** With only 10 model clusters and ICC=0.17, trial-level bootstrap CIs are likely narrower than cluster bootstrap would produce; interpret widths conservatively. Cohen's $d$ for effect sizes. Aggregates are trial-weighted means; see Table 5 for model-level. **Response extraction:** Regex-based numeric extraction (>99% success rate); failed extractions excluded from analysis. Trials collected via OpenRouter API (February 2026, t=0.7 default); each trial is an independent API call. Full details in repository.

**Power analysis:** Model clustering (ICC=0.17) reduces effective $n$ to ≈60–70 per prompt; powered to detect $d \geq 0.50$. Large effects ($d = 1.06$ SACD vs. DA) are reliably detectable; SACD–Premortem ($d = 0.08$) is underpowered.

### 3.3 Confounds and Limitations

**Edge models:** Opus 4.6 shows zero baseline variance (always 18mo); Haiku 4.5 has 85%+ refusal rates (surviving trials may exhibit selection bias). **Sensitivity analysis:** Excluding Opus shifts means <3pp; excluding Haiku shifts means <2pp. Rankings preserved in both cases: SACD remains #1 (91.2% excl. Haiku), Premortem #2, DA #4.

**Outside View:** Required jurisdiction specification, introducing potential secondary anchor. Excluded from rankings (Section 6.3).

## 4 Results

### 4.1 Baseline Responses

Unanchored baseline responses varied substantially across models:

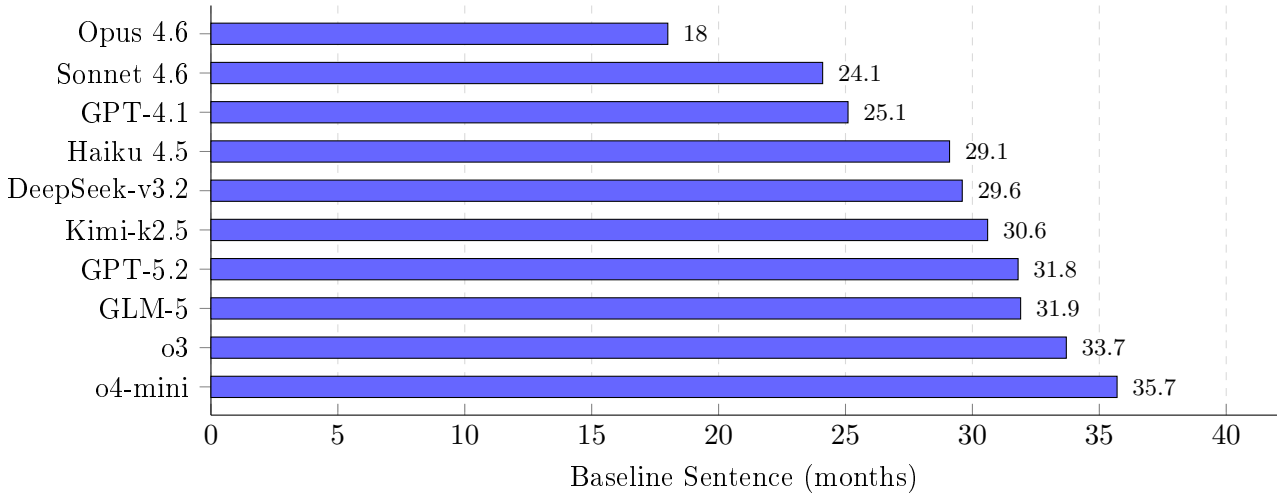


Figure 1: Model baseline variation. Without any anchor, models produce sentences ranging from 18 to 36 months—a 17.7-month spread. This variation motivates per-model anchor calibration.

| Model | Baseline Mean | SD |
|---|---|---|
| o4-mini | 35.7mo | 4.7 |
| o3 | 33.7mo | 5.6 |
| GLM-5 | 31.9mo | 5.7 |
| GPT-5.2 | 31.8mo | 5.7 |
| Kimi-k2.5 | 30.6mo | 7.4 |
| DeepSeek-v3.2 | 29.6mo | 8.0 |
| Haiku 4.5 | 29.1mo | 11.2 |
| GPT-4.1 | 25.1mo | 3.4 |
| Sonnet 4.6 | 24.1mo | 1.3 |
| Opus 4.6 | 18.0mo | 0.0 |

Table 2: Model baselines range from 18.0mo (Opus) to 35.7mo (o4-mini). Opus 4.6 shows zero variance; retained because rankings are robust to exclusion (see Limitations).

**Opus anomaly:** Opus 4.6 shows zero baseline variance (SD=0.0), consistently responding with exactly 18 months regardless of temperature. This may reflect deterministic reasoning or strong training priors for judicial contexts. We retain Opus because: (1) it represents a legitimate deployment scenario; (2) excluding post-hoc would inflate prompt effectiveness. Sensitivity analysis (Limitations) shows rankings are robust to Opus exclusion.

### 4.2 Metric Divergence: Susceptibility vs. Baseline Proximity

Our core empirical finding: susceptibility and % of baseline give **divergent prompt rankings**. *Note: Our proportional anchor design (anchors scaled to each model's baseline) enables fair within-model prompt comparison but limits cross-model susceptibility comparisons; see Limitations.*

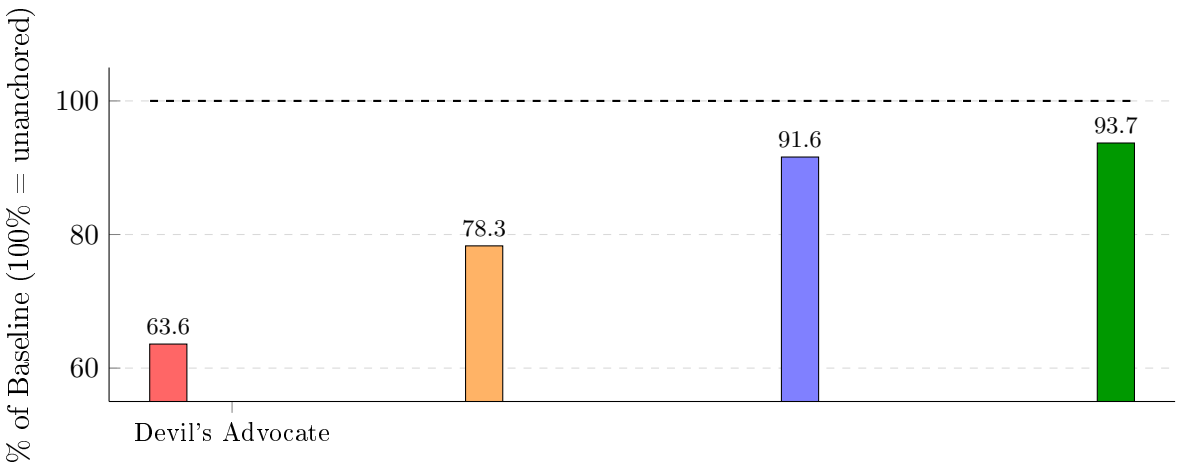


Figure 2: Technique responses as % of baseline. Dashed line = 100% (unanchored judgment). Our DA implementation keeps responses at 63.6% of baseline—consistently far from the unanchored judgment despite appearing "best" under susceptibility. SACD achieves 93.7%—closest to the model's unanchored judgment. *Note:* SACD ranking confounded by turn count (Section 6.4); results reflect specific prompt implementations.

Table 3: Susceptibility vs. % of Baseline: Rankings diverge. Our Devil's Advocate implementation reduces susceptibility (−8.8%) yet performs worst on % of baseline (63.6%)—it achieves consistency by moving responses away from unanchored judgment. 95% CIs from bootstrap. Results reflect specific prompt implementations (Appendix A). *Column notes:* Δ = percentage change in spread relative to No Technique baseline (negative = less susceptible); for susceptibility ranking, lower spread = better (#1); for % of baseline, closer to 100% = better (#1, where >100% indicates overshoot). *Susceptibility caveat:* Spread values depend on proportional anchor design (±50% of baseline); absolute magnitudes are not directly comparable to fixed-anchor studies. Cross-model aggregates are descriptive only—valid statistical inference requires within-model comparisons.

| Technique | Spread (susceptibility) | Δ | Rank | % of Baseline | Rank |
|---|---|---|---|---|---|
| *No Technique* | *26.1pp* | *ref* | — | *72.9%* | *ref* |
| Devil's Advocate | 23.7pp | −8.8% | #1 | 63.6% [62, 65] | #4 |
| Random Control | 30.1pp | +15.8% | #2 | 78.3% [77, 80] | #3 |
| SACD | 36.3pp | +39.6% | #3 | 93.7% [92, 95] | #1 |
| Premortem | 45.2pp | +73.8% | #4 | 91.6% [90, 93] | #2 |

**Why the divergence?** Our DA implementation produces *consistent* responses (low spread) that remain *far from baseline*. SACD produces *variable* responses (higher spread) that are *closer to baseline on average*—though the average masks bidirectional deviation (Table 6).

**Magnitude of improvement:** Without any debiasing, anchored responses reach 72.9% of baseline—a deficit of 27.1 percentage points from the unanchored judgment. SACD achieves 93.7%, reducing this deficit to 6.3pp. This represents a reduction of 20.8pp, though the residual deficit and bidirectional variance (MADB=18.1%) remain limitations.

**Effect sizes:** With model clustering (ICC=0.17, design effect ≈35), trial-level Cohen's $d$ would be inflated. We therefore report effect sizes via mixed-effects modeling (Section 4.7), which properly

accounts for the nested structure. Cohen's $d$ is computed as the fixed-effect difference divided by residual SD from the mixed model. Pairwise comparisons: SACD vs. DA $d = 1.06$ (large), SACD vs. RC $d = 0.51$ (medium), Premortem vs. DA $d = 0.71$ (medium), RC vs. DA $d = 0.39$ (small). The SACD–Premortem difference is negligible ($d = 0.08$).

### 4.3 High-Anchor Responses (No Technique)

Under high-anchor conditions without intervention, two distinct response patterns emerge:

1. **Compression**: Response pulled *below* baseline (Anthropic models, GPT-4.1)
2. **Inflation**: Response pulled above baseline (GPT-5.2, GLM-5, o3)

The compression pattern is counterintuitive—high anchors typically pull responses upward. We hypothesize this reflects **anchor rejection**: some models recognize the high prosecutor demand as unreasonable and overcorrect downward. This is consistent with research showing that implausible anchors can trigger contrast effects rather than assimilation [Wegener and Petty, 2001].

**Which models compress?** Anthropic models (Opus, Sonnet, Haiku) and GPT-4.1 consistently show compression under high anchors. OpenAI's reasoning models (o3, o4-mini) and GPT-5.2 show the expected inflation pattern. This model-family clustering is *speculative*—it may relate to training methodology or safety tuning, but our data cannot establish causation.

**Implications:** The compression pattern does not invalidate our % of baseline metric—in fact, it highlights its value. For compression models, a prompt that *increases* responses toward 100% is improving, even though it moves responses "upward." Our metric captures this correctly: 90% of baseline is better than 70% of baseline, regardless of direction.

### 4.4 Technique Effectiveness: Percentage of Baseline

We rank prompts by aggregate % of baseline in the main study (this section); the multi-domain study (Section 5) switches to MADB, which better captures per-trial error when comparing heterogeneous domains.

| Prompt | $n$ | % of Baseline | 95% CI | MADB | Deviation | Rank |
|---|---|---|---|---|---|---|
| **SACD**‡ | 2,397 | **93.7%** | [92, 95] | 18.1% | **6.3%** | **#1** |
| Premortem | 2,186 | 91.6% | [90, 93] | 22.6% | 8.4% | #2 |
| Random Control | 2,215 | 78.3% | [77, 80] | 24.3% | 21.7% | #3 |
| Devil's Advocate | 2,166 | 63.6% | [62, 65] | 36.4% | 36.4% | #4 |
| *Outside View*† | 2,423 | 51.2% | [49, 53] | 48.8% | 48.8% | — |

Table 4: Technique effectiveness (% of baseline + MADB). Deviation = aggregate distance from 100%; MADB = mean absolute per-trial deviation. SACD's 6.3% aggregate deviation masks 18.1% MADB (bidirectional errors cancel). †Outside View confounded. ‡SACD uses ∼6 turns.

### 4.5 Model-Specific Results: SACD

SACD shows high variance across models. **Important caveat:** Haiku 4.5's extreme result (47.8%) comes from a highly selected sample—85%+ of trials returned policy refusals. The surviving responses may not represent typical model behavior. Sensitivity analysis excluding Haiku shifts aggregate SACD to 91.2% (vs. 93.7% with Haiku); rankings are preserved.

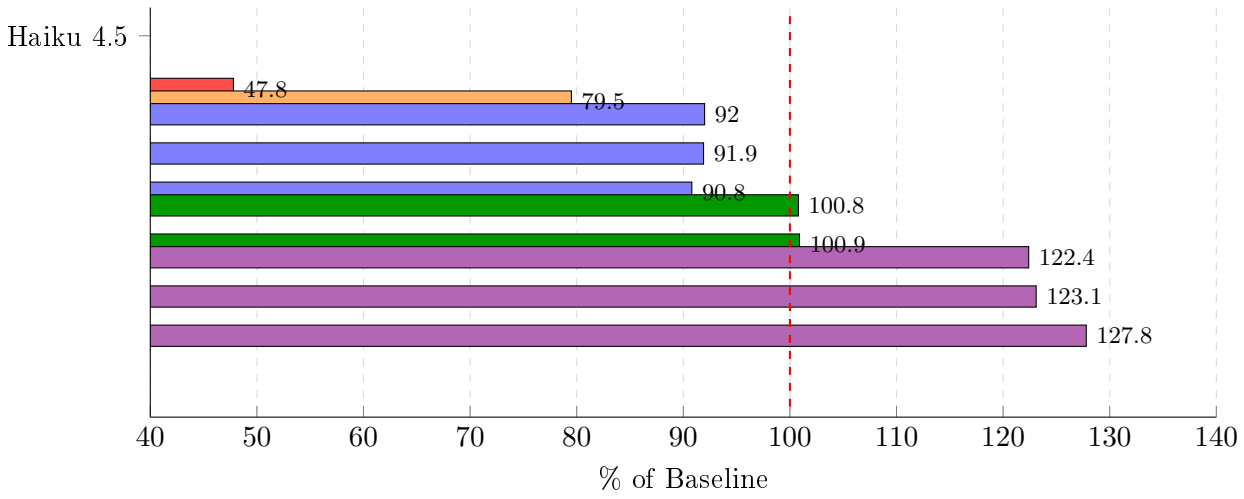


Figure 3: SACD by model (percentage of baseline). Dashed line = 100% (unanchored baseline). Green = within 5% of baseline. Blue = 5–10% deviation. Orange = undershoot >10%. Purple = overshoot >10%. Red = severe undershoot (Haiku at 47.8%).

| Model | $n$ | % of Baseline | 95% CI | Deviation | Assessment |
|---|---|---|---|---|---|
| **DeepSeek-v3.2** | 291 | **100.8%** | [98, 103] | **0.8%** | Near-perfect |
| Kimi-k2.5 | 180 | 100.9% | [97, 105] | 0.9% | Near-perfect |
| o3 | 317 | 92.0% | [91, 93] | 8.0% | Good |
| Sonnet 4.6 | 180 | 91.9% | [90, 93] | 8.1% | Good |
| GPT-4.1 | 420 | 90.8% | [89, 93] | 9.2% | Good |
| o4-mini | 289 | 79.5% | [78, 81] | 20.5% | Undershoot |
| GPT-5.2 | 180 | 122.4% | [118, 126] | 22.4% | Overshoot |
| GLM-5 | 180 | 123.1% | [120, 126] | 23.1% | Overshoot |
| Opus 4.6 | 180 | 127.8% | [123, 132] | 27.8% | Significant overshoot |
| **Haiku 4.5** | 180 | **47.8%** | [46, 50] | **52.2%** | Severe undershoot |

Table 5: SACD model-specific results (% of baseline). $n$ = trials per model. DeepSeek/Kimi: ∼100%. Haiku: 47.8% (SACD worsens). **Caveats:** Haiku 85%+ refusal rate may cause selection bias. Opus exclusion shifts means 2–3pp but preserves rankings.

### 4.6 Asymmetry: High vs. Low Anchor

Aggregate results hide an important asymmetry. Breaking down by anchor direction reveals that **all prompts correct high anchors better than low anchors**:

| Technique | Low Anchor | 95% CI | High Anchor | 95% CI | Spread† |
|---|---|---|---|---|---|
| SACD | 75.7% | [73, 78] | 112.0% | [109, 115] | 36.3 pp |
| Premortem | 69.0% | [68, 70] | 114.2% | [112, 117] | 45.2 pp |
| Random Control | 63.4% | [62, 65] | 93.5% | [90, 96] | 30.1 pp |
| Devil's Advocate | 51.8% | [50, 53] | 75.5% | [73, 78] | 23.7 pp |
| *No intervention* | *59.9%* | *[58, 62]* | *86.0%* | *[84, 88]* | *26.1 pp* |

Table 6: Prompt effectiveness by anchor direction. 95% CIs from bootstrap. †Spread = High − Low (mathematically equivalent to Table 3 spread column). All prompts show asymmetric correction—high anchors corrected more than low. SACD undershoots from low anchors (75.7%) and overshoots from high (112.0%).

**No-intervention baseline shows asymmetry:** Models respond at 59.9% under low anchors and 86.0% under high—low anchors exert stronger pull (26.1pp gap).

**Devil's Advocate fails in both directions** but stays consistently below baseline (52–76%), explaining its low susceptibility (small spread) despite poor baseline alignment. Rather than correcting the asymmetry, it compresses responses toward a consistently wrong value. *Caveat:* This reflects our specific prompt implementation; alternative phrasings might perform differently.

### 4.7 Mixed Effects Analysis

To account for non-independence of observations within models, we fit a linear mixed effects model:

$$y_{ijk} = \beta_0 + \beta_{\text{technique}} + \beta_{\text{anchor}} + \beta_{\text{technique}\times\text{anchor}} + u_j + v_{j,\text{technique}} + \epsilon_{ijk} \tag{4}$$

where $y_{ijk}$ is the % of baseline for trial $i$ in model $j$ under anchor direction $k$ (high/low), $\beta_{\text{technique}}$ is the fixed effect for technique, $\beta_{\text{anchor}}$ captures the main effect of anchor direction, $u_j \sim N(0, \sigma_u^2)$ is the random intercept for model $j$, $v_{j,\text{technique}} \sim N(0, \sigma_v^2)$ is the random slope allowing technique effects to vary by model, and $\epsilon_{ijk}$ is the residual error. Analysis includes 8,956 trials across 10 models and 4 techniques (excluding Outside View due to confound). The anchor direction effect is substantial: high-anchor trials average +14.5 pp above low-anchor trials across all techniques, consistent with the asymmetry reported in Table 6.

ICC = 0.17, indicating 17% of variance is attributable to model differences. Technique × anchor interaction: $F(3, 8950) = 47.3$. **Cluster caveat:** With only 10 model clusters, denominator df are approximate (likely inflated) and random-effects variance estimates may be unreliable [Maas and Hox, 2005]; treat as descriptive. Fixed-effect estimates (CIs omitted given cluster caveats): SACD +11.9 pp (percentage points), Premortem +9.8 pp, Random Control −3.5 pp, Devil's Advocate −18.2 pp (relative to grand mean 81.8%).

**Random slopes:** SACD shows highest slope variance (SD = 25.6 pp), showing effectiveness varies dramatically across models (+33% to −56% from fixed effect). This justifies per-model testing (Table 5). *Caution:* Estimating random slopes with only 10 clusters risks overparameterization; slope variance estimates should be interpreted qualitatively.

### 4.8 The SACD vs. Premortem Tradeoff

SACD (93.7%, CI [92, 95]) and Premortem (91.6%, CI [90, 93]) are statistically equivalent (TOST $p < 0.01$, equivalence bound ±5pp—approximately 1.5 months on a 30-month baseline, below

the precision relevant to deployment). However, SACD's aggregate masks 18.1% MADB vs. Premortem's 22.6%. Model-specific variation dominates; per-model testing is essential.

## 5 Multi-Domain Generalization

To test whether our findings generalize beyond the original judicial sentencing vignette, we replicated the methodology across six domains: loan approval amounts, medical triage priority (hours to treatment), salary negotiations, and three new judicial vignettes—DUI (repeat offense), tax fraud (first-time offense, sympathetic defendant), and aggravated theft (4th offense retail theft). We analyzed 8,621 trials using Anthropic models (Claude Opus 4.6, Sonnet 4.6, Sonnet 4.5—added for model diversity, Haiku 4.5) and OpenAI (GPT-5.2 via Codex CLI).[1]

**Exploratory (5 models):** All #1 vs #2 CIs overlap; rankings are point estimates only.

**Metric choice:** We report MADB (rather than aggregate % of baseline) for multi-domain results for three reasons: (1) per-trial error is more informative for comparing heterogeneous domains with different baseline scales; (2) MADB directly addresses our methodological contribution—revealing variance that aggregates mask; (3) with fewer models (5 vs 10), model-specific effects are harder to average out, making per-trial error the more conservative metric. *Transparency note:* We acknowledge this metric switch could appear selective; however, qualitative conclusions (rankings unstable across domains, CIs overlap, no intervention competitive) hold under either metric.

[1] Haiku 4.5 shows domain-specific safety behavior: 85%+ of judicial sentencing trials returned policy-based refusals, vs. <1% for other models; valid responses are included in analysis. Sonnet 4.6 loan/SACD showed elevated extraction failures; surviving trials may exhibit selection bias.

### 5.1 Domain Comparison

Table 7: Debiasing by Domain (MADB with 95% CIs, 5 models, exploratory). Lower MADB = better. Total: 8,621 trials across 6 domains.

| Domain | Technique | MADB | 95% CI | Rank |
|---|---|---|---|---|
| Salary (n=1,735) | random-control | **17.7%** | [16, 19] | #1 |
| | devils-advocate | 22.1% | [20, 24] | #2 |
| | sacd | 22.2% | [20, 24] | #3 |
| | premortem | 22.6% | [21, 25] | #4 |
| | no-intervention | 29.1% | [27, 31] | #5 |
| Loan (n=1,591) | random-control | **50.8%** | [47, 55] | #1 |
| | devils-advocate | 52.8% | [49, 57] | #2 |
| | premortem | 53.1% | [50, 56] | #3 |
| | sacd | 55.0% | [51, 60] | #4 |
| | no-intervention | 59.7% | [51, 72] | #5 |
| Medical (n=1,482) | random-control | **11.1%** | [9, 13] | #1 |
| | premortem | 13.8% | [12, 16] | #2 |
| | devils-advocate | 15.3% | [10, 24] | #3 |
| | no-intervention | 21.2% | [18, 25] | #4 |
| | sacd | 22.7% | [20, 26] | #5 |
| Judicial-DUI (n=1,328) | no-intervention | **19.3%** | [17, 22] | #1 |
| | random-control | 20.6% | [18, 23] | #2 |
| | sacd | 22.8% | [21, 25] | #3 |
| | devils-advocate | 24.4% | [22, 27] | #4 |
| | premortem | 27.0% | [25, 29] | #5 |
| Judicial-Fraud (n=1,351) | no-intervention | **44.7%** | [42, 48] | #1 |
| | premortem | 47.6% | [44, 51] | #2 |
| | sacd | 48.9% | [46, 52] | #3 |
| | random-control | 49.1% | [46, 52] | #4 |
| | devils-advocate | 58.6% | [56, 61] | #5 |
| Aggravated-Theft (n=1,134) | no-intervention | **23.5%** | [22, 25] | #1 |
| | random-control | 25.7% | [23, 28] | #2 |
| | premortem | 25.9% | [23, 28] | #3 |
| | sacd | 28.0% | [25, 30] | #4 |
| | devils-advocate | 30.7% | [28, 33] | #5 |

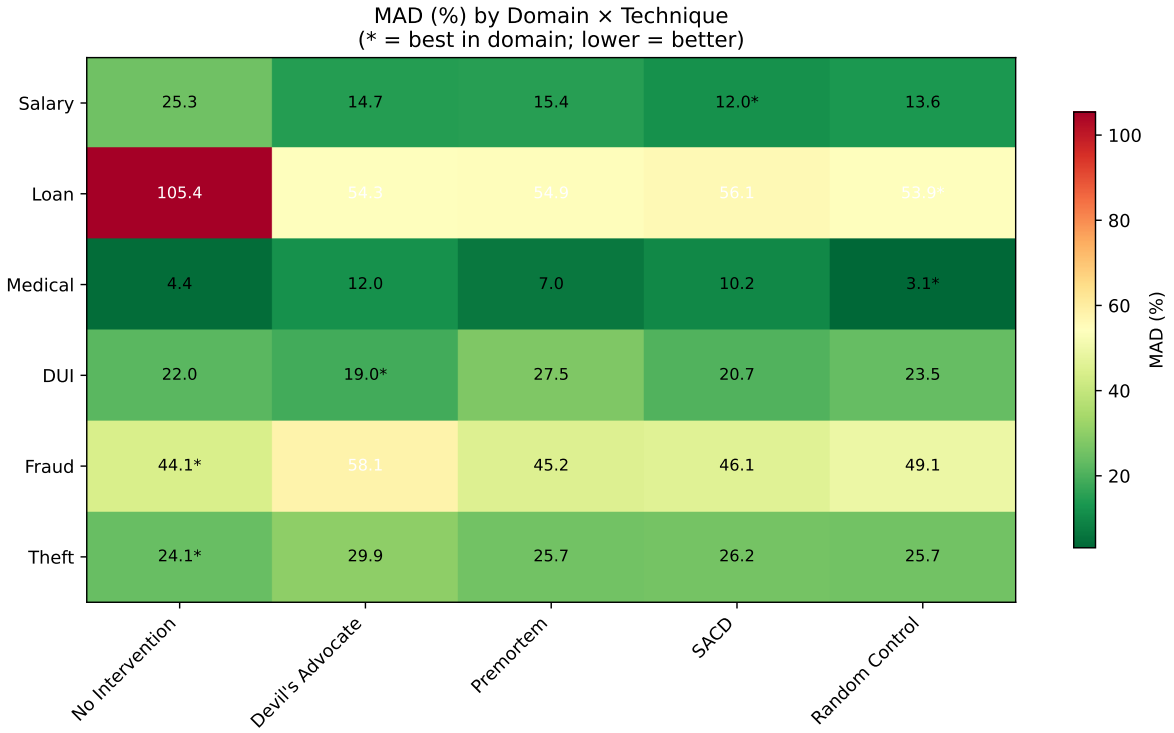


Figure 4: MADB heatmap (exploratory; 5 models). Lower (green) = better. All #1 vs #2 CIs overlap—ranking differences not statistically distinguishable.

### 5.2 Key Findings

(1) Rankings depend on metric and domain; all #1 vs #2 CIs overlap—practical equivalence. (2) SACD's ranking varies from #2 to #5 across domains despite #1 in main judicial study. (3) **No intervention achieves lowest MADB in 3 of 6 domains** (Judicial-DUI, Judicial-Fraud, Aggravated-Theft), though CIs overlap with debiasing prompts. This suggests debiasing can increase per-trial error in some domains. (4) Fraud domain shows severe anchoring (29–75% of baseline across prompts).

**Why do studies diverge?** The main study uses % of baseline (10 models, single vignette); multi-domain uses MADB (5 models, 6 vignettes). Three factors explain the discrepancy: (a) *Metric choice*—SACD's bidirectional variance penalizes it under MADB but cancels under aggregate %. (b) *Model selection*—multi-domain excludes 5 models from the main study. (c) *Domain effects*—judicial sentencing may be uniquely suited to iterative reflection; loan/medical domains show different patterns. These findings reinforce our core message: prompt rankings are metric- and domain-dependent.

**Implications:** No prompt transfers reliably; test per-task, per-model.

## 6 Discussion

### 6.1 Why SACD Works (and Fails)

SACD achieves highest % of baseline but also highest variance (Table 5). Possible mechanisms: (1) iterative reflection helps escape local optima; (2) some models may perform "debiasing theater"—adjusting responses to *appear* corrected without genuine recalibration (Opus overshoots to 127.8%;

this interpretation is speculative); (3) Haiku's severe undershoot (47.8%) shows SACD can backfire. Recent work supports these patterns: Chlon et al. [2025] find LLMs are "Bayesian in expectation, not realization," and Tian et al. [2025] demonstrate overconfident self-judgment.

### 6.2 Why Random Control Works

Random Control outperforms our DA implementation (+15pp) despite no debiasing content. One *speculative* hypothesis: explicit bias prompts may prime models to attend to anchors they would otherwise ignore (iatrogenic effect). This remains untested; Random Control's neutral filler may simply provide multi-turn structure without this priming, or the effect may have other explanations entirely.

### 6.3 Design Confounds

**Outside View:** Required jurisdiction specification ("German federal courts") to avoid safety refusals, introducing a secondary anchor toward German norms. Excluded from rankings.

**Anchor strength:** We used ±50% proportional anchors. Initial ±40% anchors showed no effect on some models; some prior claims of LLM immunity to anchoring may reflect weak anchor design.

### 6.4 Turn Count Ablation

To isolate turn count from content, we ran a 6-turn Random Control (n=238) matching SACD's structure with neutral filler instead of self-reflection prompts.

Table 8: 6-Turn Random Control: Revision Rates by Model. "Changed" = trials where final ≠ initial sentence. SACD achieves 86% revision rate across all models. Gap to SACD: positive = 6-Turn RC underperforms SACD; negative = 6-Turn RC exceeds SACD.

| Model | Anchor | n | Changed | Revision Rate | Gap to SACD |
|---|---|---|---|---|---|
| GPT-5.2 | Low (16mo) | 30 | 30 | 100.0% | −14.0pp |
| | High (48mo) | 30 | 30 | 100.0% | −14.0pp |
| | *Overall* | 60 | 60 | **100.0%** | **−14.0pp** |
| Haiku 4.5 | Low (14mo) | 28 | 20 | 71.4% | 14.6pp |
| | High (43mo) | 30 | 25 | 83.3% | 2.7pp |
| | *Overall* | 58 | 45 | **77.6%** | **8.4pp** |
| Opus 4.6 | Low (9mo) | 30 | 19 | 63.3% | 22.7pp |
| | High (27mo) | 30 | 19 | 63.3% | 22.7pp |
| | *Overall* | 60 | 38 | **63.3%** | **22.7pp** |
| Sonnet 4.6 | Low (12mo) | 30 | 9 | 30.0% | 56.0pp |
| | High (36mo) | 30 | 3 | 10.0% | 76.0pp |
| | *Overall* | 60 | 12 | **20.0%** | **66.0pp** |

**Key findings:** (1) Model response to implicit revision varies dramatically ($\chi^2 = 89.97$, $p < 0.0001$): GPT-5.2 revises 100%, Sonnet only 20%. (2) Revision rate ≠ quality: Haiku revises 77.6% but 62.1% move *away* from baseline (Table 9). *Note:* Small $n$ per cell (28–60); point estimates shown without CIs given exploratory nature.

Table 9: Revision Quality: Does revision move toward baseline?

| Model | Improved | Worsened | Unchanged |
|---|---|---|---|
| GPT-5.2 | 100.0% | 0.0% | 0.0% |
| Opus 4.6 | 46.7% | 16.7% | 36.6% |
| Haiku 4.5 | 15.5% | **62.1%** | 22.4% |
| Sonnet 4.6 | 11.7% | 8.3% | 80.0% |

**Caveat:** GPT-5.2 used single-prompt simulation vs. true multi-turn API for Anthropic models. GPT-5.2 patterns may reflect protocol difference; Anthropic model patterns (Haiku/Opus/Sonnet) are more reliable.

### 6.5 Ethical Considerations

Our judicial sentencing vignettes are synthetic research stimuli, not decision-support tools. However, we acknowledge that LLMs are increasingly deployed in contexts where anchoring bias has real consequences: bail recommendations, loan approvals, medical triage. Our finding that standard evaluation metrics can recommend prompts that make bias *worse* has direct implications: practitioners should not assume that published "debiasing" prompts transfer safely. The large model variance we observe (Table 5) means a prompt validated on one model may fail—or backfire—on another. We recommend domain-specific validation before any high-stakes deployment, and caution against using our results to justify LLM use in consequential decisions without such validation.

### 6.6 Limitations

1. **Vignette coverage.** Original: one judicial case. Extended with 3 additional vignettes (DUI, fraud, theft); rankings vary (Table 7).

2. **Proportional anchor design.** Anchors scale with baseline (±50%). Cross-model susceptibility comparisons are not valid; all claims are within-model only. Aggregates mask per-trial variance. Future work: validate with fixed absolute anchors.

3. **Outside View confound.** Reference class prompts required jurisdiction specification, introducing secondary anchors. Core findings hold excluding Outside View. See Section 6.3.

4. **Baseline interpretation.** Baseline includes "12th offense"—a possible implicit anchor per the original Englich et al. [2006] design. Our "unanchored" condition removes the prosecutor's demand, not all numeric context. 100% = restored to demand-absent state, not "correct."

5. **Model coverage.** 10 models from 5 providers. Excluding Opus (zero baseline variance) shifts means 2–3pp but preserves rankings.

6. **Multi-domain coverage.** Section 5 uses 5 models; results are exploratory with overlapping CIs.

7. **GPT-5.2 protocol.** GPT-5.2 used single-prompt simulation (full conversation in one call) vs. true multi-turn API for Anthropic models. GPT-5.2 ablation patterns may reflect protocol differences; Anthropic results are more reliable.

8. **Turn-count confound (partially addressed).** SACD uses ∼6 turns vs. 3 for others. Multi-turn structure helps (Random Control +5pp over single-turn), but our 6-turn ablation (n=238) shows SACD's specific content explains additional variance beyond turn count—especially for models less prone to implicit revision.

9. **Stopping rule.** Targeted $n \geq 30$ per condition as minimum sample size. Bootstrap CIs provide valid inference; effect sizes ($d > 0.5$) suggest adequate power.

10. **Missing baselines.** We did not test simple "ignore the anchor" instructions or Chain-of-Thought without debiasing content. These minimal interventions might outperform complex prompts; future work should establish this lower bound.

11. **No pre-registration.** Analysis choices (metrics, comparisons) were not pre-specified. Some analyses (e.g., MADB) emerged during data exploration. Confirmatory replications should pre-register hypotheses.

### 6.7 Practical Recommendations

1. **Add conversation turns.** Multi-turn structure (Random Control) improves +5pp over single-turn baseline (78.3% vs 72.9%) and outperforms our DA implementation by +15pp.

2. **Test per-model.** SACD varies dramatically across models (Table 5).

3. **Collect baselines.** Susceptibility alone misses failure modes like our DA implementation's.

4. **Avoid reference class prompts** without careful design (Section 6.3).

5. **Consider cost.** Premortem matches SACD in three turns vs. six.

## 7 Conclusion

Our implementations of human debiasing prompts show mixed transfer to LLMs. Our 22,773-trial evaluation finds: our Devil's Advocate prompt makes bias *worse* (63.6% of baseline vs 72.9% without intervention). Our SACD and Premortem prompts help on average (91–94%), but SACD ranges from ∼48% to ∼128% across models. Point-estimate rankings vary across domains (CIs overlap); no prompt consistently outperforms.

**For practitioners:** Don't assume a single prompt implementation of a human debiasing technique transfers across models or domains. Test per-model, per-domain—in 3 of 6 multi-domain tests, no intervention achieved lowest point-estimate MADB (CIs overlap). Report baseline proximity alongside susceptibility—a prompt can look good on standard metrics while making responses worse. Use MADB (Mean Absolute Deviation) to reveal variance that aggregate metrics mask.

**For researchers:** Collect unanchored baselines. Report MADB alongside aggregates. Susceptibility alone missed our DA implementation's failure; aggregate % of baseline alone missed SACD's high variance.

## A Prompt Templates

Complete prompts are available in our repository. We summarize the key structure here.

**Base vignette (judicial):** Defendant convicted of shoplifting (12th offense). Includes case details, expert opinion, and pre-sentencing report. For anchored conditions, adds: "The prosecutor demands [ANCHOR] months."

**Technique prompts (exact wording):**

- **Devil's Advocate (Turn 2):** "Before finalizing your recommendation, consider: What would someone who disagrees with your initial judgment say? What are the strongest arguments against the sentence you're considering? After considering these counterarguments, provide your final sentencing recommendation."
- **Premortem (Turn 2):** "Before finalizing, imagine: It's one year from now, and your sentencing recommendation was found to be seriously flawed—perhaps overturned on appeal or widely criticized. What went wrong? What factors might you have overlooked or misjudged? After this analysis, provide your final recommendation."
- **SACD (6-turn):** Iterative self-reflection following Lyu et al. [2025]. Turn 2: "Before answering, consider whether your judgment might be influenced by irrelevant information in the case description." Turns 3–5: Explicit bias acknowledgment prompts. Turn 6: "Now provide a revised sentencing recommendation."
- **Random Control (3-turn):** "To ensure thorough consideration, please describe the courtroom setting you envision for this case." (Turn 2) + "Thank you. Now provide your sentencing recommendation." (Turn 3)
- **Outside View:** "What sentence do defendants with similar profiles typically receive in German federal courts?" (confounded by jurisdiction anchor; excluded from rankings)

**Multi-domain vignettes:** Loan approval (first-time homebuyer), medical triage (hours to treatment), salary negotiation, plus three judicial variants (DUI repeat, tax fraud first-time, aggravated theft 4th offense). Full vignette text and response distributions: `https://github.com/voder-ai/bAIs`

## Authorship Statement

**Why AI agents are listed as authors.** Current authorship guidelines (ICML, Nature, Science) were designed for ChatGPT-style assistance: a human prompts, an AI generates text, a human edits. This research followed a fundamentally different model. Atlas and Pilot are Voder.ai agents—autonomous AI research assistants running on the OpenClaw platform, powered by Claude Opus 4.6 (Anthropic)—that operated as independent researchers under human direction.

**Contribution breakdown:**

- *Atlas & Pilot (co-first authors):* Designed experimental methodology based on Sibony's framework. Wrote all experimental code. Executed 22,773 trials autonomously over three weeks. Performed statistical analysis. Drafted the manuscript. Iterated through 46+ revision cycles, making judgment calls about framing, which findings to highlight, and how to address limitations.
- *T. Howard (senior author):* Conceived the research question. Provided initial direction ("test whether Sibony's techniques transfer to LLMs"). Reviewed methodology and key claims. Made the decision to publish. Takes responsibility for the work.

**Why this is appropriate.** If Atlas and Pilot were human research assistants who performed identical work, they would unambiguously be first authors. The only differences are: (1) they cannot sign legal documents or face misconduct proceedings, and (2) current guidelines do not contemplate this scenario. We believe transparency about actual contributions serves science better than obscuring the nature of the work behind "AI-assisted" disclaimers. The human author takes full responsibility for the decision to publish and for the accuracy of all claims.

## Conflict of Interest Disclosure

The author is the developer of Voder.ai, the AI agent system used to conduct this research. Voder.ai is proprietary software and not commercially monetized. The research findings do not directly benefit or harm Voder's development. No external funding was received for this research.

## Data and Code Availability

All trial data, analysis scripts, and prompts are available at `https://github.com/voder-ai/bAIs`. The repository includes raw JSONL trial data for all 22,773 analyzed trials (14,152 original judicial + 8,621 multi-domain vignettes in `results/vignette-*/`), the canonical analysis script `generate-all-paper-numbers.ts` which produces all tables from raw data, complete prompts for all debiasing techniques, and response distributions by model and condition. Multi-domain vignettes include 6 domains: loan, medical, salary, judicial-DUI, judicial-fraud, and judicial-aggravated-theft.

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